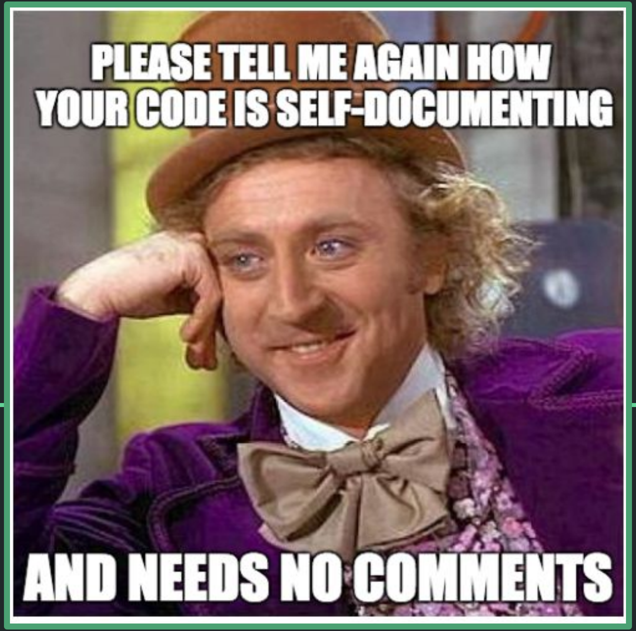
Documentation & Collaboration

BILD 62

Goals for today

- Define guidelines for writing and documenting good code
- Describe & practice code review

Writing good code
and documenting it



How do we write good code for humans?

- **Use good structure**
  - If you design your program using separate functions for each task, avoid copying + pasting (functions and loops instead), and consider structure beforehand, you'll be set up for success
- **Use good naming**
- **Use code comments and include documentation**

Ideas: Shannon Ellis (COGS 18)

What does this code do?

```python
def ff(jj):
    oo = list(); jj = list(jj)
    for ii in jj: oo.append(str(ord(ii)))
    return '+'.join(oo)
ff('Hello World.')
```

clear function name

some comments

separate lines

```python
def return_unicode(input_list):
    string = list() # []
    input_list = list(input_list)

    for character in input_list:
        string.append(str(ord(character)))

    output_string = '+'.join(string)
    return output_string

return_unicode('Hello World.')
```

better names!

Ideas: Shannon Ellis (COGS 18)

Writing useful comments

- Good code has good documentation - but code documentation should *not* be used to try and fix unclear names, or bad structure.
- Rather, comments should add any additional context and information that helps explain what the code is, how it works, and why it works that way.
    - focus on the how and why, over literal 'what is the code'
    - explain any context needed to understand the task at hand
    - give a broad overview of what approach you are taking to perform the task
    - if you're using any unusual approaches, explain what they are, and why you're using them

Ideas: Shannon Ellis (COGS 18)

Types of comments

Block comments

```
# this box describes block
# comments and the best way
# to write them
```

- apply to some (or all) code that follows them
- are indented to the same level as that code.
- Each line of a block comment starts with a # and a single space

Inline comments `# inline`

- to be used sparingly
- to be separated by at least two spaces from the statement
- start with a # and a single space

Ideas: Shannon Ellis (COGS 18)

Docstrings are in-code text that describe modules, classes and functions. They describe the operation of the code.

- Available to you outside of the source code using `help()`
- Stored as the `__doc__` attribute and can be accessed from there too
- Common structure:
  - Starts and ends with triple quotes
  - One sentence overview at the top - the task/goal of function
  - **Parameters** : description of function arguments, keywords & respective types
  - **Returns** : explanation of returned values and their types

More info: PEP 257 – Docstring Conventions | peps.python.org

Ideas: Shannon Ellis (COGS 18)

```python
# Let's fix this code!
def convert_to_unicode(input_string):
    """Converts an input string into a string containing the unicode code points.

    Parameters
    ----------
    input_string : string
        String to convert to code points

    Returns
    -------
    output_string : string
        String containing the code points for the input string.
    """

    output = list()
    # Converting a string to a list, to split up the characters of the string
    input_list = list(input_string)

    for character in input_list:
        temp = ord(character)
        output.append(temp)

    output_string = '+'.join(output)

    return output_string
```

Now with a docstring!

Ideas: Shannon Ellis (COGS 18)

What should be included in a docstring?

- A brief overview sentence about the code
- Input arguments/parameters and their types
- Returned variables and their types

Style guides

- **Coding style** refers to a set of conventions for how to write good code.
- Python Enhancement Proposals (PEPs) are written by the people responsible for the Python Programming language and propose standards for writing Python code
- PEP8 is an accepted proposal that outlines the style guide for Python.

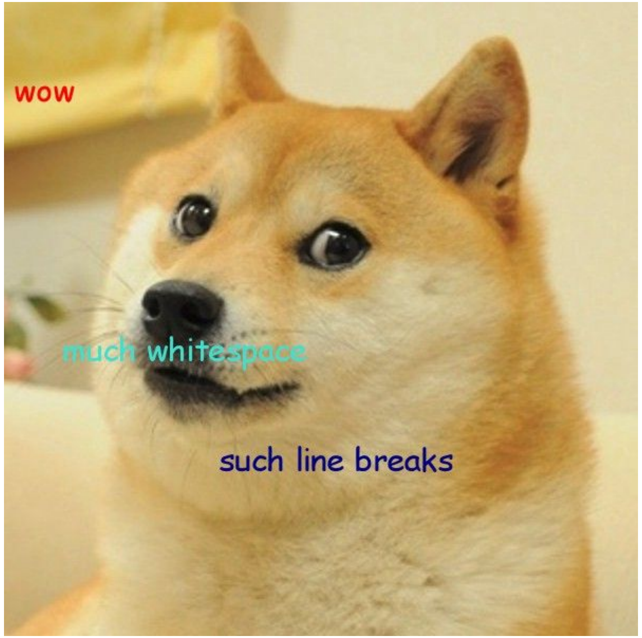
https://peps.python.org/pep-0008/

Naming conventions

- `CapWords` (leading capitals, no separation) for Classes
- `snake_case` (all lowercase, underscore separator) for variables, functions, and modules

Spacing conventions

- Put one (and only one) space between each element
- Index and assignment don't have a space between opening & closing '()' or '[]'
- One statement per line
- Blank line conventions:
  - Use 2 blank lines between functions & classes, and 1 between methods
  - Use 1 blank line between segments to indicate logical structure



Goals for today

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Code review is a process for systematically reviewing someone else's code.

[Code Reviews Explained [+ Why They Matter] | Atlassian]

Negative criticism usually fails in one or more the following ways

1. **It isn't strategic.** The critic does not think about what specifically they want to change or what goals and solutions they can offer.
2. **It isn't improvement oriented.** The critic doesn't make suggestions as to how to improve.
3. **It attacks self-esteem.** The critic uses labels (such as "lazy"), speaks in absolutes, and does not allow the recipient to save face.
4. **It uses the wrong words.** The critic uses negative statements and words like "should" instead of "could."
5. **It comes with no supporting evidence.** Critic does not support comments with evidence or fair comparisons.

https://www.science.org/content/article/joy-criticism

Categories for code review

1. Functionality: Does the code work as intended? Is it robust? If not, where is the issue?

2. Documentation & Style: Is the code properly documented and commented? Where should the documentation be better? Does spacing & capitalization follow PEP-8 guidelines?

3. Error handling: Does the code handle errors properly? Where could the error-handling be better?

4. Other suggestions: Are there areas where the code could be more concise? Functions that could have been used, but weren't?

DOCUMENTATION & STYLE (35)			
Docstrings *(10 points)*	There are docstrings on functions & classes that describe 1) code's function 2) parameters/inputs, 3) returns	Functions and classes contain docstrings that partially describe the function/class	Functions/classes do not contain docstrings
Documentation *(10 points)*	Any line of code that is not immediately self-explanatory to someone with a BILD62 level of Python knowledge is explained via inline or block comments and/or Markdown text	There are some block or inline comments that helpfully explain the code	Very few block or inline comments throughout project
Code Style *(15 points)*	There is good code layout - spacing between code segments, etc; names used are descriptive and follow naming conventions	Spacing is somewhat inconsistent and sometimes not following conventions, names are vague	Spacing is inconsistent throughout & multiple style guide rules are not followed

Relevant parts of final project rubric

Regardless of how you collaborate, in the end, everyone needs to submit via **DataHub** AND **Canvas**

so that we can test your code

so that we can leave comments

No need for data files;
ONLY ONE PER GROUP!

PLEASE INCLUDE YOUR DATA FILES!
Only one per group is fine

Today: review our “CodeReview” notebook in the four categories & submit on Canvas.

(As you work on your project, review each other’s code!)